Question 1

Not answered

Marked out of 1.00

Flag question

Find:

$$(01010 \oplus 10101) \oplus 01000$$

Answer: 10111



Question 2

Not answered

Marked out of 1.00

Flag question

Given two propositions:

p = "I visit Hanoi."

q = "I visit Ho Chi Minh mausoleum"

Which sentence on the left corresponds to the expression on the right?

I went to Hanoi, but I did not visit Ho Chi Minh mausoleum.	Choose...	p^-q
I visit Ho Chi Minh mausoleum only if I go to Hanoi.	Choose...	q->p
I cannot visit Ho Chi Minh mausoleum if I do not go to Hanoi	Choose...	-p -> -q
Whenever I go to Hanoi, I visit Ho Chi Minh mausoleum.	Choose...	p -> q

Question 3

Not answered

Marked out of 1.00

Flag question

Which proposition is logically equivalent to

$$(p \rightarrow q) \vee [\neg p \rightarrow (q \vee r)] \text{ ?}$$

-p V q V p V (q V r

T T T

Select one:

c

a. $r \vee q$

b. $p \vee q$

c. T

d. $q \rightarrow r$

e. $p \vee q \vee r$

Question 4

Not answered

Marked out of 1.00

Flag question

Let *p*, *q* be two propositions. Which propositions are logically equivalent to $p \rightarrow q$?

$\neg q \rightarrow p$	Choose...	false
$\neg p \rightarrow \neg q$	Choose...	false
$\neg p \rightarrow q$	Choose...	false
$\neg q \rightarrow \neg p$	Choose...	true



Question 5

Not answered

Marked out of 1.00

Flag question

Let $P(x)$ be a propositional function with domain $\{-1, 0, 1\}$

Which proposition on the left has the same truth value as the proposition on the right?

$\forall x((x \neq 1) \rightarrow \neg P(x))$ — Choose... -P(-1) ^ -P(0)

$\exists x \neg P(x)$ — Choose... -P(-1) V -P(0) V -P(1)

$\exists x((x \neq 1) \wedge P(x))$ — Choose... P(-1) V P(0)

Question 6

Not answered

Marked out of 1.00

Flag question

Translate the logical expression into sentence, domain is all real numbers

$$\forall x \forall y(((x<0) \wedge (y<0)) \rightarrow (xy>0))$$

Select one:

- a. For each negative number x there is a negative number y such that xy is positive
- b. There is a negative number x and there is a negative number y such that xy is positve
- c. The product of two negative numbers is negative
- d. The product of two negative numbers is positive

Question 7

Not answered

Marked out of 1.00

Flag question

Given an argument:

"If Jack is a soccer player then Jack is rich. Jack is not rich. Therefore Jack is not a soccer player."

Choose correct statement:

Select one:

- a. This argument is a fallacy
- b. This valid argument is based on disjunctive syllogism
- c. This valid argument is based on modus tollens
- d. This valid argument is based on hypothetical syllogism
- e. This valid argument is based on modus ponens

Question 8

Not answered

Marked out of 1.00

Flag question

Find the negation of

$$\forall x \forall y(\exists z \neg T(x,y,z) \wedge Q(x,y))$$

Select one:

- a. $\exists x \exists y(\forall z T(x,y,z) \vee \neg Q(x,y))$.
- b. $\forall x \exists y(\forall z T(x,y,z) \vee Q(x,y))$.
- c. $\forall x \exists y(\forall z \neg T(x,y,z) \wedge \neg Q(x,y))$.
- d. $\exists x \exists y(\forall z \neg T(x,y,z) \vee Q(x,y))$.



Question 9

Not answered

Marked out of 1.00

Flag question

Let A ={1, 2, 4, 6, 7, 9, 8} B = {3, 1, 5, 7, 6}. Which set has the maximum cardinality?

Select one:

a. $A-B$

b. $B-A$

c. $A \cap B$



Question
10

Not answered

Marked out of 1.00

Flag question

Let A={0, a}, B={0, b}. Find the cardinality of the set P(A x B).

Answer: 16



Question
11

Not answered

Marked out of 1.00

Flag question

Compute $\left\lfloor \frac{3}{2} - \left\lceil 3 + \frac{4}{5} \right\rceil \right\rfloor$

Answer: -3



Question
12

Not answered

Marked out of 1.00

Flag question

Let f(X) = 5X + 4, g(X) = 4X + 3. Suppose that f o g (X) = aX + b.
Find a + b.

Answer: 39



Question
13

Not answered

Marked out of 1.00

Flag question

Compute $\left\lfloor \frac{3}{2} - \left\lceil 3 + \frac{5}{4} \right\rceil \right\rfloor$

Answer: -4



Question
14

Not answered

Marked out of 1.00

Flag question

Let f: Z x Z --> Z, f(m, n) = n+1. Choose correct answer:

Select one:

- a. f is one-to-one but not onto
- b. f is onto but not one-to-one
- c. f is a bijection
- d. f(x) is neither one-to-one nor onto



Question
15

Not answered

Marked out of 1.00

Flag question

Compute

$$\sum_{i=10}^{20} i^2.$$

Answer: 2585

